Beyond the Dyad: Linguistic Style Transfer Across Conversational Partners

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People can pick up ways of speaking from a conversational partner. This project asks whether those habits carry over when they write to someone else. Kronmüller and Guerra [1] distinguished adaptation to individual speakers from adaptation to the wider conversation in a reference task. In a controlled production task, Ostrand and Ferreira [2] found that participants' syntax shifted with the forms they heard across partners. We extend this question to several dimensions of style in naturally interleaved conversations. We predict that after receiving messages from A, a player P will write to B in a style closer to A's, even after accounting for P's earlier usage and the messages B has sent to P.

The public *Diplomacy* corpus [3] contains 17,289 private messages from 12 seven-player games. We replay each game in message order, predicting the style of each outgoing message from the sender, recipient, and prior conversation. Once observed, the message becomes part of the recipient's history. To measure style, we mask place names and move orders, then extract a standardized vector of features such as pronoun use, hedges, contractions, and punctuation. Paired rewrites will test whether the vector captures changes in expression when content stays fixed and remains stable when only the content changes [4, 5]. Our primary analysis considers the whole vector, with individual features examined separately.

A hierarchical Gaussian model tracks what a player learns from incoming messages, drawing on Hawkins et al.'s account of shared and partner-specific expectations and Xu and Futrell's model of how linguistic preferences update with experience [6, 7]. To predict a message, each model blends the player's early writing style with the style estimate it uses for that recipient. A weight fitted to observed messages determines the size of the shift. The no-adaptation baseline fixes that weight at zero. If A's style changes while B's history stays fixed, a partner-specific learner predicts a shift only in messages to A. Full pooling predicts comparable shifts to both partners under matched conditions. Partial pooling allows A's influence to carry over to B, with a smaller shift when P has already received substantial input from B.

In our pilot, stylistic associations were clearer within a partnership, where the input also tended to be more recent. We will compare inputs at similar distances in the message sequence while accounting for game phase, message length, and reply context. Messages unavailable to P provide a negative control for shared circumstances. In simulations, we vary A's input with B's history fixed and redistribute identical inputs across partners to separate exposure from source. On games held out one at a time, higher predictive log scores for models with shared learning and shifts toward A in messages to B would support cross-partner adaptation. The observational results will help identify which predictions to test under controlled exposure.

References

[1] Kronmüller, E., & Guerra, E. (2026). Adapting pragmatic interpretation in multiparty conversations: Mutual exclusivity inferences across speaker and context. *Language and Cognition, 18*, e43. doi:10.1017/langcog.2026.10086

[2] Ostrand, R., & Ferreira, V. S. (2019). Repeat after us: Syntactic alignment is not partner-specific. *Journal of Memory and Language, 108*, 104037. doi:10.1016/j.jml.2019.104037

[3] Peskov, D., Cheng, B., Elgohary, A., Barrow, J., Danescu-Niculescu-Mizil, C., & Boyd-Graber, J. (2020). It takes two to lie: One to lie, and one to listen. *Proceedings of ACL 2020.* aclanthology.org/2020.acl-main.353/

[4] Halvani, O., & Graner, L. (2021). POSNoise: An effective countermeasure against topic biases in authorship analysis. *Proceedings of ARES 2021.* doi:10.1145/3465481.3470050

[5] Wegmann, A., Schraagen, M., & Nguyen, D. (2022). Same author or just same topic? Towards content-independent style representations. *Proceedings of RepL4NLP 2022.* aclanthology.org/2022.repl4nlp-1.26/

[6] Hawkins, R. D., Franke, M., Frank, M. C., Goldberg, A. E., Smith, K., Griffiths, T. L., & Goodman, N. D. (2023). From partners to populations: A hierarchical Bayesian account of coordination and convention. *Psychological Review, 130*(4), 977-1016. doi:10.1037/rev0000348

[7] Xu, W., & Futrell, R. (2024). A hierarchical Bayesian model for syntactic priming. *Proceedings of CogSci 2024.* arXiv:2405.15964